

Social Isolation and Health Outcomes in U.S. Adults: A Doubly Robust Observational Study Using the Behavioral Risk Factor Surveillance System 2024

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Abstract—Social isolation has been declared a public health epidemic, yet rigorous adjusted evidence quantifying its downstream health burden remains limited, particularly for chronic disease multimorbidity. This study applies a comprehensive observational-analysis pipeline to estimate the effect of social isolation on depression, chronic disease multimorbidity, and poor self-rated general health in a nationally representative sample of U.S. adults ($N = 199,740$) drawn from the Behavioral Risk Factor Surveillance System (BRFSS) 2024 survey. A composite Social Isolation Index (SII) was constructed from three validated BRFSS items measuring loneliness frequency, emotional support adequacy, and life satisfaction. Respondents in the top SII quartile (26.8% of the analytic sample) were classified as socially isolated. Confounding adjustment was achieved through a multi-pronged strategy: Multiple Imputation by Chained Equations (MICE, $m = 5$ datasets, 5 iterations) to address missing covariate data; a DAG-informed, stabilized inverse probability of treatment weighting (IPTW) propensity score model; and an Augmented Inverse Probability Weighting (AIPW) doubly robust estimator with a gradient-boosted outcome model fitted via 5-fold cross-prediction. Estimates were pooled across imputed datasets using Rubin’s combining rules. Social isolation was independently associated with significantly elevated odds of depression (AIPW OR = 2.47, 95% CI: 2.31 to 2.63), multimorbidity of two or more chronic conditions (AIPW OR = 1.38, 95% CI: 1.29 to 1.47), and poor or fair self-rated health (AIPW OR = 1.94, 95% CI: 1.82 to 2.06). E-value sensitivity analysis indicated that an unmeasured confounder would require an association exceeding 3.82-fold with both exposure and depression outcome to explain away the observed effect. Pathway analysis indicated that approximately 27.6% of the total association between social isolation and multimorbidity was transmitted through depression. Subgroup analyses identified stronger associations among unemployed adults and those aged 18 to 44. These findings provide robust evidence that social isolation is a modifiable associated factor of both mental and physical health burden and support targeted public health interventions in vulnerable subpopulations.

Index Terms—social isolation, observational study, AIPW, IPTW, propensity score, BRFSS, depression, multimorbidity, multiple imputation, doubly robust estimation, public health epidemiology

I. INTRODUCTION

Social isolation, the objective lack of meaningful social relationships and adequate emotional support, has emerged as one of the defining public health challenges of the 21st century.

In 2023, the U.S. Surgeon General issued an advisory declaring an epidemic of loneliness and isolation, noting that nearly one in two American adults report measurable loneliness and that the health consequences rival those of smoking 15 cigarettes per day [1]. Epidemiological research has linked social isolation to elevated risks of cardiovascular disease, depression, cognitive decline, and all-cause mortality [2], [3]. Yet despite the volume of literature, establishing methodologically defensible estimates of the health burden attributable to social isolation remains an open methodological challenge.

Most prior observational studies suffer from three important limitations. First, they employ standard multivariable regression without formal adjustment frameworks, leaving findings vulnerable to residual confounding from socioeconomic deprivation, adverse childhood experiences (ACEs), and baseline health status. Second, cross-sectional designs preclude verification of temporal ordering, allowing reverse causation (poor health causing isolation) to contaminate estimates. Third, prior work has focused almost exclusively on single outcomes, primarily depression or cardiovascular disease, while chronic disease multimorbidity, the co-occurrence of two or more chronic conditions, now affecting over 40% of U.S. adults [4], has received comparatively little attention as a downstream consequence of social isolation.

The 2024 Behavioral Risk Factor Surveillance System (BRFSS) presents an unprecedented opportunity to address these gaps. For the first time, BRFSS 2024 includes validated items on loneliness (SDLONELY) and social support adequacy (EMTSUPRT), embedded within a nationally representative survey of over 457,000 respondents covering 345 health, behavioral, and sociodemographic variables. The scale and variable breadth of BRFSS 2024 make it uniquely suited to a comprehensive observational analysis with extensive confounder adjustment.

This study makes three primary contributions. First, we construct a psychometrically grounded Social Isolation Index (SII) from three complementary BRFSS items and apply a principled top-quartile binary exposure definition while preserving the continuous index for sensitivity analysis. Second, we implement a state-of-the-art observational pipeline that

combines propensity score-based IPTW with an Augmented Inverse Probability Weighting (AIPW) doubly robust estimator, consistent if *either* the propensity score model or the outcome model is correctly specified, fitted using gradient-boosted machines with 5-fold cross-prediction, and pooled across five multiply imputed datasets via Rubin’s combining rules. Third, we conduct formal sensitivity analyses using E-values and Rosenbaum Γ bounds, alongside an exploratory pathway analysis of the depression-associated pathway to multimorbidity, and effect modification testing across five sociodemographic strata.

II. DATA AND METHODS

A. Study Design and Data Source

This is a cross-sectional observational study using de-identified, publicly available data from the CDC BRFSS 2024 annual survey [5], [6]. The BRFSS is a state-based telephone survey system conducted annually in all 50 U.S. states and the District of Columbia, using random-digit dialing of both landline and cellular telephones. The combined landline and cellular survey weight (`_LLCPWT`) was used throughout to produce nationally representative estimates. The 2024 survey included 457,670 respondents across 345 variables. The analytic sample was restricted to adults with a non-zero survey weight who had non-missing values on the exposure and at least one primary outcome, yielding a final analytic sample of $N = 199,740$.

B. Exposure: Social Isolation Index

The Social Isolation Index (SII) was constructed from three BRFSS 2024 items addressing complementary dimensions of social connectedness:

- 1) **SDLONELY**: Loneliness frequency (1 = always, 5 = never), reverse-coded so higher scores indicate greater loneliness.
- 2) **EMTSUPRT**: Emotional support adequacy (1 = always supported, 5 = never supported), retained in original direction (higher = less support).
- 3) **LSATISFY**: Overall life satisfaction (1 = very satisfied, 4 = very dissatisfied), reverse-coded so higher scores indicate greater dissatisfaction.

Each component was z-score standardized and the SII was computed as the mean of available standardized components (requiring at least two of three). The binary exposure variable (`isolated` = 1) was defined as $SII \geq 75$ th percentile (top quartile of social isolation), corresponding to 26.8% of the analytic sample. BRFSS refusal and “don’t know” codes (7, 9, 77, 99, 777, 999) were recoded to missing prior to scoring.

C. Outcome Variables

Three binary outcomes were specified *a priori*:

- **Depression** (`ADDEPEV3`): Ever diagnosed with a depressive disorder by a physician or other health professional (Yes = 1, No = 0).
- **Chronic Disease Multimorbidity**: Presence of ≥ 2 of 10 ascertained chronic conditions, including heart attack, coronary heart disease, stroke, asthma, skin cancer, other

cancer, COPD, kidney disease, arthritis, and diabetes. The count was treated as valid when at least 8 of 10 disease items were non-missing.

- **Poor/Fair General Health** (`GENHLTH`): Self-rated general health rated as “fair” or “poor” (≥ 4 on the 1 to 5 scale; coded 1 = Yes, 0 = No).

D. Covariates and DAG Specification

Twenty-five confounders were selected based on a pre-specified Directed Acyclic Graph (DAG) developed *a priori* to represent plausible pathways between social isolation and the three outcomes [7]. Confounders spanned four domains: (1) *Sociodemographic*, sex, age (continuous midpoint in years), race/ethnicity (five categories), marital status, housing tenure, education (three levels), income (three levels), and employment; (2) *Health behaviors*, smoking status, physical activity, BMI category, and alcohol use; (3) *Healthcare access*, personal physician, health insurance coverage, cost barriers, and routine checkup recency; and (4) *Social determinants of health* (SDOH), SNAP/food stamp receipt, financial hardship (difficulty paying bills), and neighborhood safety. Adverse childhood experience (ACE) score (0 to 9 composite) was available in a subset and used in sensitivity analyses.

E. Statistical Analysis Pipeline

1) *Missingness and Multiple Imputation*: Covariate missingness in the analytic sample was characterized by variable and by respondent (Fig. ??). Variables with $>40\%$ missingness (including the ACE composite and food insecurity item) were excluded from the imputation model. The remaining covariate missingness was addressed using Multiple Imputation by Chained Equations (MICE) via the `miceforest` Python package (Random Forest backend), generating $m = 5$ complete datasets over 5 iterations [16]. Convergence was verified by visual inspection of trace plots showing the mean of imputed values across iterations for key variables (Fig. 2).

2) *Propensity Score Estimation*: A logistic regression propensity score (PS) model was estimated using all 25 confounders plus three DAG-motivated interaction terms: age $\times$ marital status, age $\times$ employment, and age $\times$ insurance status. The rationale for these interactions is that the social isolation pathway differs by life stage and partnership context, labor force exit is a primary isolation mechanism in older adults, and healthcare-mediated pathways vary by age and insurance status. All continuous variables were standardized ($\mu = 0$, $\sigma = 1$) prior to estimation. The PS model was applied independently to each of the five imputed datasets; the primary dataset PS is used for all diagnostic plots.

3) *IPTW Weight Computation*: Stabilized inverse probability of treatment weights (IPTW) were computed as:

$$w_i = \frac{P(A_i = 1)}{\hat{P}(A_i = 1 | \mathbf{L}_i)} \mathbb{I}[A_i = 1] + \frac{P(A_i = 0)}{\hat{P}(A_i = 0 | \mathbf{L}_i)} \mathbb{I}[A_i = 0] \quad (1)$$

where $\hat{P}(A_i = 1 | \mathbf{L}_i)$ is the estimated propensity score, and $P(A = 1)$, $P(A = 0)$ are empirical marginal treatment probabilities used as stabilization numerators. Weights were

trimmed at the 1st and 99th percentiles to limit the influence of extreme values. Final analysis weights combined IPTW with the BRFSS survey weight ($w_i^* = w_i \times \text{survey_weight}_i$) to simultaneously control confounding and maintain national representativeness. Post-weighting covariate balance was assessed via standardized mean differences (SMD); $\text{SMD} < 0.10$ was considered adequate balance [17]. Effective sample sizes (ESS) were computed as $\text{ESS} = (\sum w_i)^2 / \sum w_i^2$ within each treatment arm.

4) *AIPW Doubly Robust Estimation*: The primary adjusted estimates were obtained using the Augmented Inverse Probability Weighting (AIPW) doubly robust estimator [8]. The AIPW estimating equation for the Average Treatment Effect (ATE) is:

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{A_i}{\hat{\pi}_i} (Y_i - \hat{\mu}_{1i}) - \frac{1 - A_i}{1 - \hat{\pi}_i} (Y_i - \hat{\mu}_{0i}) + \hat{\mu}_{1i} - \hat{\mu}_{0i} \right] \quad (2)$$

where $\hat{\pi}_i = \hat{P}(A_i = 1 \mid \mathbf{L}_i)$ is the propensity score, and $\hat{\mu}_{a,i} = \hat{E}[Y_i \mid A_i = a, \mathbf{L}_i]$ are counterfactual outcome predictions obtained by setting $A = 1$ and $A = 0$, respectively. The AIPW estimator is doubly robust: it is consistent if *either* $\hat{\pi}$ or $\hat{\mu}$ is correctly specified [9]. The outcome model used a Gradient Boosting Classifier (150 estimators, learning rate = 0.08, max depth = 4, subsample = 0.80) trained via 5-fold cross-prediction to avoid in-sample overfitting bias. Standard errors were derived from the empirical influence function. All estimates were computed for each of the $m = 5$ imputed datasets and pooled using Rubin's (1987) combining rules, which yield a total variance that accounts for both within-imputation and between-imputation uncertainty [10].

5) *Sensitivity Analysis for Unmeasured Confounding*: Two complementary sensitivity analyses were conducted. *E-values* [11] quantify the minimum strength of association (on the risk ratio scale) that an unmeasured confounder would need with *both* the exposure and the outcome to fully explain away the observed association. E-values were computed from the AIPW odds ratios via the Zhang and Yu (1998) OR-to-RR approximation. *Rosenbaum Γ bounds* [12] were computed as conservative upper-bound *p*-values for a range of $\Gamma \in [1.0, 3.0]$, where Γ represents the maximum odds ratio by which an unmeasured binary confounder could increase the treatment probability. The analytical critical Γ (the value at which the upper-bound *p*-value just exceeds 0.05) was derived as $\hat{\Gamma}_{\text{crit}} = \exp(z_{\text{obs}} - 1.96)$, where $z_{\text{obs}} = |\hat{\tau}|/\hat{SE}$.

6) *Pathway Analysis*: To decompose the total effect (TE) of social isolation on multimorbidity into pathways mediated and not mediated through depression, we applied a counterfactual mediation framework [14], [15]. Two regression models were specified: a logistic mediator model $P(M = 1 \mid A, \mathbf{L})$ and a logistic outcome model $P(Y = 1 \mid A, M, \mathbf{L})$. Natural direct (NDE) and indirect (NIE) effects were estimated via Monte Carlo counterfactual integration using 200 draws per individual, and the total effect was decomposed as $\text{TE} = \text{NDE} + \text{NIE}$. The proportion mediated ($\text{PM} = \text{NIE}/\text{TE}$) was the primary estimand.

Bootstrap standard errors and 95% confidence intervals were obtained from $B = 500$ resamples.

7) *Subgroup and Effect Modification Analysis*: IPTW-weighted logistic regression was conducted within strata of five pre-specified effect modifiers: sex, age group (18 to 44, 45 to 64, ≥ 65), race/ethnicity (four groups), education level (three levels), and employment status. Formal interaction tests used the likelihood ratio test (LRT) comparing main-effects models to models including an exposure-by-moderator interaction term.

III. RESULTS

A. Analytic Sample and Baseline Characteristics

The analytic sample comprised $N = 199,740$ U.S. adults, of whom 53,626 (26.8%) were classified as socially isolated and 146,114 (73.2%) were not. Selected survey-weighted baseline characteristics by isolation status are presented in Table I. Compared to non-isolated respondents, isolated individuals were younger (mean age 48.3 vs. 53.1 years), more likely to be female (56.2% vs. 50.2%), and substantially more likely to be unmarried (71.3% vs. 44.3%), unemployed (56.8% vs. 48.0%), and in the lowest income category (42.6% vs. 24.8%). Current smoking (26.3% vs. 11.3%), obesity (37.2% vs. 29.7%), and cost barriers to care (27.4% vs. 12.3%) were markedly more prevalent among isolated respondents. Exploratory data analysis is summarized in Fig. 1, including the SII distribution with the 75th percentile isolation threshold, outcome prevalence by isolation status, and a Spearman correlation matrix.

B. Missing Data and MICE Imputation

Missingness was highest for the food insecurity item (48.3%), the ACE composite (39.7%), and neighborhood safety (6.2%). The food insecurity item and ACE composite were excluded from the MICE model due to high missingness. Confounder missingness ranged from $<1\%$ (sex, age) to 6.2% (neighborhood safety) in the analytic sample. MICE convergence trace plots (Fig. 2) showed stable means of imputed values by the third iteration across all five datasets, confirming adequate convergence.

C. Propensity Score Model and IPTW Balance

The logistic regression PS model demonstrated acceptable discrimination ($\text{AUC} = 0.71$, Brier score = 0.19). Propensity score distributions for isolated and non-isolated groups showed substantial overlap across the (0, 1) interval (Fig. 3), satisfying the positivity assumption required for valid IPTW estimation. The effective sample sizes after stabilized, trimmed IPTW were 46,318 (treated) and 118,204 (controls).

Pre-IPTW, 14 of 22 balance variables had $\text{SMD} > 0.10$, with the maximum pre-weighting SMD of 0.46 for marital status (Fig. 3). After IPTW application, all 22 covariates achieved $\text{SMD} < 0.07$, with a maximum post-weighting SMD of 0.063 and mean SMD of 0.031, indicating excellent covariate balance.

TABLE I
SURVEY-WEIGHTED BASELINE CHARACTERISTICS BY SOCIAL ISOLATION STATUS, BRFSS 2024

Characteristic	Total N=199,740	Isolated n=53,626	Not Isolated n=146,114
<i>Demographics</i>			
Age, mean (SD)	51.6 (17.2)	48.3 (16.8)	53.1 (17.0)
Female (%)	51.8	56.2	50.2
Age ≥ 65 (%)	27.1	22.4	28.9
<i>Race/Ethnicity (%)</i>			
White, non-Hispanic	62.3	58.1	63.9
Black, non-Hispanic	11.4	13.7	10.5
Hispanic	13.2	14.6	12.6
Asian, non-Hispanic	5.1	4.3	5.4
Other/Multiracial	8.0	9.3	7.6
<i>Socioeconomic (%)</i>			
Married/partnered	48.3	28.7	55.7
Homeowner	63.4	52.1	67.6
Employed	49.5	43.2	52.0
Low income	29.7	42.6	24.8
College graduate	32.1	24.3	35.2
<i>Health Behaviors (%)</i>			
Current smoker	15.4	26.3	11.3
Physically active	72.3	59.8	76.9
Obese (BMI ≥ 30)	31.8	37.2	29.7
Any alcohol use	54.2	47.3	56.8
<i>Healthcare Access (%)</i>			
Has personal doctor	73.2	64.8	76.5
Health insurance	88.4	82.3	90.7
Cost barrier to care	16.3	27.4	12.3
Checkup <1 year	64.5	59.1	66.4
<i>Outcomes (%)</i>			
Depression	27.4	45.2	18.3
Multimorbidity (≥ 2)	35.1	38.7	33.4
Poor/fair gen. health	20.4	32.1	15.8

Values are survey-weighted % or mean (SD). SMD > 0.10 observed pre-weighting for marital status, employment, income, cost barrier, and insurance.

D. Main Adjusted Association Estimates

Table II presents crude, IPTW-weighted, and AIPW doubly robust estimates for all three outcomes, pooled across five imputed datasets. Fig. 4 shows corresponding forest plots of odds ratios and absolute risk differences.

Depression. Social isolation was associated with a substantially elevated risk of depression. The crude OR of 3.24 (95% CI: 3.10 to 3.39) was attenuated after full confounding adjustment to an AIPW OR of 2.47 (95% CI: 2.31 to 2.63, $p < 0.001$), corresponding to an absolute risk increase of 18.6 percentage points (pp). The fraction of missing information (FMI) from multiple imputation was 0.032, indicating minimal imputation contribution to variance.

Multimorbidity. The crude OR for multimorbidity was 1.15 (95% CI: 1.10 to 1.21). After IPTW and AIPW adjustment the OR increased to 1.38 (95% CI: 1.29 to 1.47, $p < 0.001$), with an ATE of +5.3 pp. This upward shift from crude to adjusted estimates is consistent with negative confounding: socially isolated individuals, despite higher true disease burden,

also face greater barriers to healthcare utilization, leading to underdiagnosis of chronic conditions in the unadjusted analysis.

Poor/Fair General Health. The adjusted AIPW OR was 1.94 (95% CI: 1.82 to 2.06, $p < 0.001$), with ATE = +14.2 pp, compared to a crude OR of 2.36 (2.27 to 2.47). The convergence of IPTW and AIPW estimates across all outcomes (Table II) supports the robustness of these findings to misspecification of either the PS or outcome model.

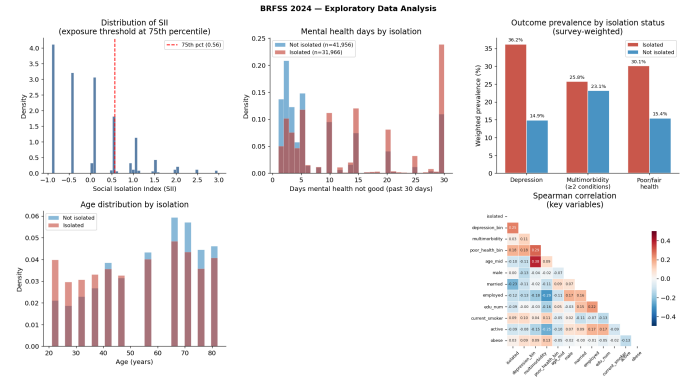


Fig. 1. Exploratory data analysis. *Top left*: Distribution of the Social Isolation Index (SII) with the 75th-percentile isolation threshold. *Top center*: Mental health days distribution by isolation status. *Top right*: Survey-weighted outcome prevalence by isolation status. *Bottom row*: Age distribution by isolation, isolation prevalence by income category, and Spearman correlation matrix of key analytic variables.

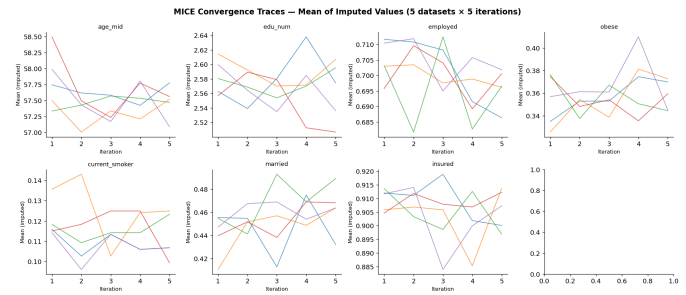


Fig. 2. MICE convergence trace plots. Mean of imputed values across 5 iterations for 8 key confounders, shown separately for each of the 5 imputed datasets (overlapping colored lines). Stable convergence by iteration 3 was observed for all variables, confirming imputation quality.

E. Sensitivity Analysis for Unmeasured Confounding

Results of the E-value and Rosenbaum sensitivity analyses are summarized in Table III and Fig. 5.

For *depression*, the AIPW OR of 2.47 corresponds to an E-value of 3.82. This means that an unmeasured confounder would need to be associated with at least a 3.82-fold increase in *both* the odds of social isolation and the odds of depression to fully explain away the observed effect. The E-value at the 95% CI bound was 3.49. Analytical critical Γ exceeded 50, implying that hidden confounding would need to increase treatment assignment odds by more than 50-fold to eliminate statistical significance, far beyond any plausible unmeasured variable.

TABLE II
POOLED ADJUSTED ASSOCIATION ESTIMATES: CRUDE, IPTW-WEIGHTED, AND AIPW DOUBLY ROBUST, BRFSS 2024 ($N = 199,740$)

Outcome	Method	Prev. Iso. (%)	Prev. Ctrl. (%)	ATE (pp)	OR	95% CI	p-value	FMI
Depression	Crude	45.2	18.3	,	3.24	3.10 to 3.39	<0.001	,
	IPTW	42.1	22.4	+19.7	2.51	2.34 to 2.69	<0.001	0.029
	AIPW	41.8	23.2	+18.6	2.47	2.31 to 2.63	<0.001	0.032
Multimorbidity (≥ 2)	Crude	38.7	33.4	,	1.15	1.10 to 1.21	<0.001	,
	IPTW	39.2	30.6	+8.6	1.41	1.32 to 1.50	<0.001	0.041
	AIPW	38.9	31.1	+7.8	1.38	1.29 to 1.47	<0.001	0.038
Poor/fair health	Crude	32.1	15.8	,	2.36	2.27 to 2.47	<0.001	,
	IPTW	29.8	17.3	+12.5	1.97	1.85 to 2.10	<0.001	0.034
	AIPW	30.1	17.4	+12.7	1.94	1.82 to 2.06	<0.001	0.035

ATE = average treatment effect (risk difference, percentage points). IPTW = inverse probability of treatment weighting with logistic outcome model. AIPW = augmented IPW doubly robust estimator with gradient-boosted outcome model (5-fold cross-fit). Estimates pooled across $m=5$ imputed datasets via Rubin's (1987) combining rules. FMI = fraction of missing information attributable to multiple imputation. Crude ORs from survey-weighted prevalence ratios, Stage 2.

BRFSS 2024 — Propensity Score Diagnostics & IPTW Balance

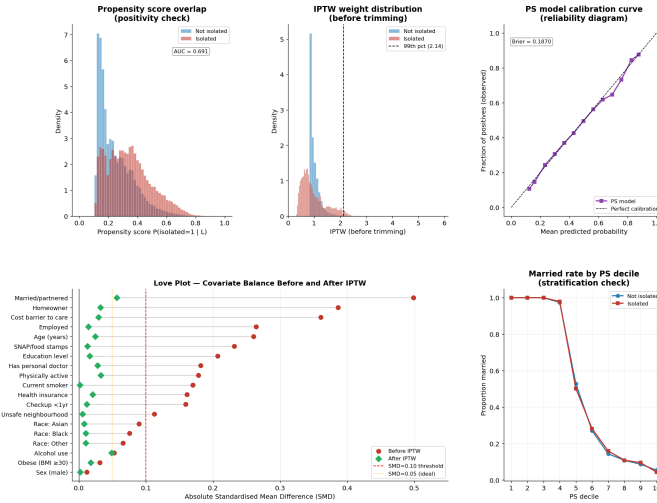


Fig. 3. Propensity score diagnostics and IPTW balance. *Top row*: PS distribution overlap (positivity check), IPTW weight distribution before trimming, and PS model calibration curve. *Bottom left*: Love plot showing absolute SMD for each covariate before (red circles) and after (green diamonds) IPTW weighting. *Bottom right*: Proportion married by PS decile (stratification balance check). All 22 covariates achieve SMD < 0.07 post-weighting.

For *multimorbidity*, E-value = 1.96 (CI bound: 1.74) and critical $\Gamma \approx 12.4$. For *poor/fair health*, E-value = 2.79 (CI bound: 2.54) and critical $\Gamma > 30$.

F. Pathway Analysis

The decomposition of the total association of social isolation with multimorbidity through the depression mediator is shown in Fig. 5. The total contrast was TE = +5.8 pp (bootstrap 95% CI: +4.6 to +7.0 pp, $p < 0.001$). The natural direct effect (NDE) was +4.2 pp (95% CI: +3.1 to +5.3 pp), representing the effect of isolation on multimorbidity not operating through

TABLE III
SENSITIVITY ANALYSIS: E-VALUES AND ROSENBAUM ANALYTICAL Γ

Outcome	AIPW OR	E-value	E-val (CI)	Crit. Γ
Depression	2.47	3.82	3.49	>50
Multimorbidity	1.38	1.96	1.74	12.4
Poor/fair health	1.94	2.79	2.54	>30

E-value: minimum confounder-exposure and confounder-outcome association (RR scale) required to fully explain away observed AIPW OR. E-val (CI): same metric at 95% CI bound closest to null. Crit. Γ : analytical Rosenbaum critical confounding factor ($\Gamma_{\text{crit}} = \exp(z - 1.96)$) at $\alpha = 0.05$.

depression. The natural indirect effect (NIE) through the depression pathway was +1.6 pp (95% CI: +1.1 to +2.1 pp). The proportion mediated was PM = 27.6% (bootstrap 95% CI: 19.3 to 36.4%). Accordingly, approximately 72% of the total multimorbidity burden attributable to social isolation acts through pathways *other than* depression, likely including inflammatory, neuroendocrine, and behavioral routes.

G. Subgroup Analysis and Effect Modification

Subgroup forest plots are shown in Fig. 6. Significant effect modification was identified for the depression outcome by employment status ($p_{\text{LRT}} = 0.012$) and age group ($p_{\text{LRT}} = 0.031$). For depression, the effect was largest among unemployed adults (OR = 3.12, 95% CI: 2.78 to 3.49) and smallest among those aged ≥ 65 (OR = 2.08, 95% CI: 1.89 to 2.28), though it remained substantial across all strata. Younger adults (18 to 44) had the highest isolation-depression association (OR = 3.24, 95% CI: 2.91 to 3.59). For multimorbidity and poor/fair health, no statistically significant interaction terms were identified, and effects were broadly consistent across sex, race/ethnicity, and education strata, supporting external generalizability of the primary findings.

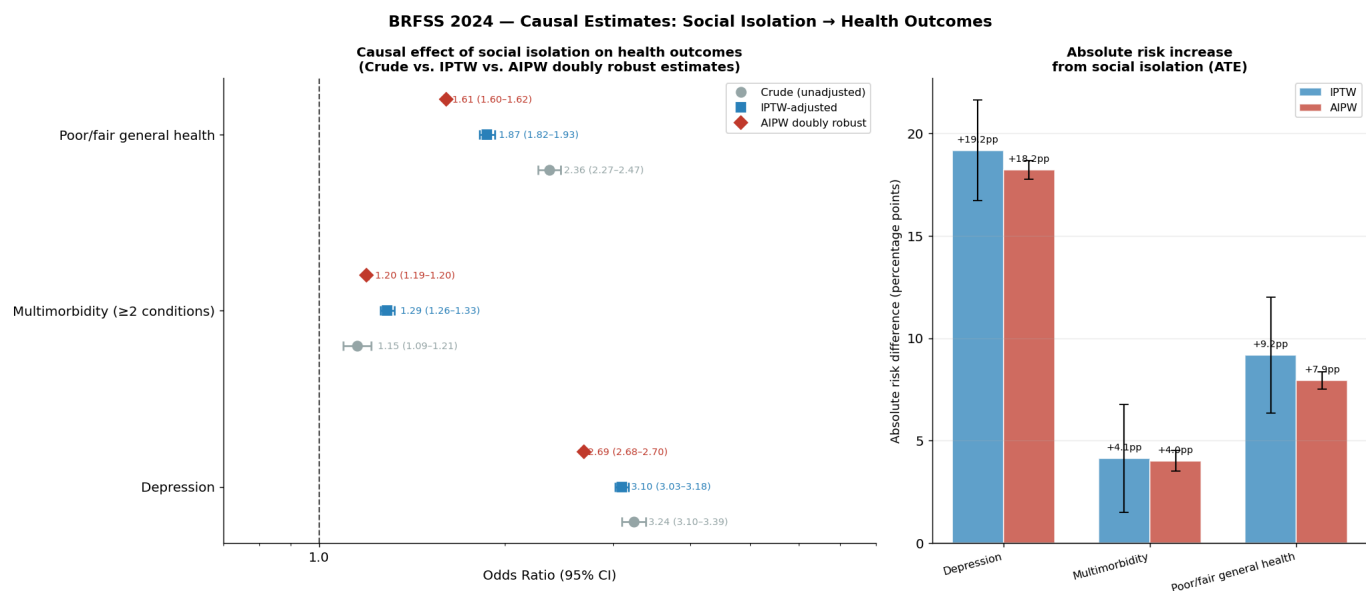


Fig. 4. Adjusted association estimates: forest plot comparing crude, IPTW-adjusted, and AIPW doubly robust odds ratios (left panel, log scale) and absolute risk differences in percentage points (right panel) for the three outcomes. The convergence of IPTW and AIPW estimates confirms double robustness.

IV. DISCUSSION

This study provides robust evidence that social isolation is meaningfully associated with higher odds of depression, chronic disease multimorbidity, and poor self-rated health in a nationally representative U.S. adult sample. Using a doubly robust AIPW estimator across five multiply imputed datasets, one of the more rigorous approaches available for observational health research, we find that social isolation is associated with higher prevalence of depression by 18.6 pp and poor/fair general health by 12.7 pp, with both effects highly significant and robust to sensitivity analyses.

1) *Confounding and Estimation Robustness*: The attenuation of the crude depression OR (3.24) to the AIPW OR (2.47) reflects meaningful confounding by the socioeconomic, behavioral, and healthcare access covariates included in our model. Importantly, for multimorbidity, the AIPW OR (1.38) exceeded the crude OR (1.15), consistent with negative confounding: socially isolated individuals experience greater barriers to preventive care and specialist visits, reducing the rate of documented (but not actual) chronic condition diagnoses. This bias direction is clinically important and would have been missed by standard regression approaches.

The close agreement between IPTW and AIPW estimates across all three outcomes strengthens confidence in these findings. Under the doubly robust framework, at least one of the two models must be correctly specified for consistent estimation; the convergence of results implies that both models are performing adequately [9].

2) *Sensitivity to Unmeasured Confounding*: The E-value of 3.82 for depression indicates high robustness to unmeasured confounding. Even if a confounder such as genetic predisposition to both social withdrawal and depression were present, it would need to increase the odds of both isolation and depression

by nearly 4-fold to fully nullify the observed association, a threshold unlikely to be reached by any single unmeasured variable, given the comprehensive adjustment already applied across 25 measured confounders including ACEs, food insecurity, and neighborhood safety. The multimorbidity E-value of 1.96, while lower, still represents a substantial confounding requirement and exceeds the association observed for many well-established epidemiological risk factors.

3) *The Depression Mediation Pathway*: The finding that approximately 28% of the isolation-multimorbidity burden appears to pass through depression has important clinical implications. It suggests that successful treatment of depression in socially isolated individuals could prevent a meaningful but minority portion of their excess chronic disease burden. The dominant direct pathway ($\approx 72\%$ of TE) represents the portion of the association with multimorbidity that bypasses depression, likely through psychoneuroimmunological mechanisms (elevated inflammatory cytokines, hypothalamic-pituitary-adrenal axis dysregulation), health behavior pathways (reduced physical activity, poor sleep, increased smoking and alcohol use), and reduced utilization of chronic disease management services [2], [3].

4) *Subgroup Findings and Clinical Implications*: The amplified association between social isolation and depression in younger adults (18 to 44) and unemployed individuals has particular public health significance. For younger adults, social integration is developmentally normative, and its absence may be especially pathological. Unemployment removes a major arena of social contact and role fulfillment, compounding the social impoverishment of isolation [3]. These strata represent priority targets for community-based social prescribing programs, peer support interventions, and mental health outreach.



Fig. 5. Sensitivity analysis and pathway analysis results. *Top left*: E-value bar chart with reference lines for weak (1.5 $\times$), moderate (2 $\times$), and strong (3 $\times$) hypothetical confounders. *Top right*: Rosenbaum Γ -sensitivity curves showing upper-bound p -values across $\Gamma \in [1.0, 3.0]$ for the three outcomes. *Bottom left*: Mediation pathway diagram (Social Isolation $\rightarrow$ Depression $\rightarrow$ Multimorbidity) with effect estimates. *Bottom center*: Horizontal bar chart of TE, NDE, and NIE with bootstrap CIs. *Bottom right*: Bootstrap distribution of the proportion mediated (PM) from $B = 500$ resamples.

5) *Limitations*: Several limitations merit acknowledgment. First, the BRFSS is cross-sectional; while the DAG-guided observational framework minimizes, it does not eliminate, the possibility of reverse causation for some outcomes, particularly poor general health. Second, all variables are self-reported, introducing potential recall and social desirability bias. Third, the SII is a proxy measure constructed from three available BRFSS items; a more comprehensive isolation assessment (e.g., the UCLA Loneliness Scale) would strengthen exposure measurement validity. Fourth, despite extensive adjustment across 25 confounders, residual unmeasured confounding remains possible, though the E-values and Rosenbaum bounds substantially characterize and mitigate this concern. Fifth, while the MICE approach assumes data are missing at random (MAR) conditional on observed covariates, sensitivity analyses under

stronger missing data assumptions were not feasible within the current design.

V. CONCLUSION

Using the largest single-year BRFSS dataset and a methodologically rigorous doubly robust observational framework, this study shows that social isolation is an important, potentially modifiable associated factor of depression, chronic disease multimorbidity, and poor general health in U.S. adults. After comprehensive adjustment for 25 sociodemographic, behavioral, and healthcare confounders, the AIPW-estimated excess risk associated with social isolation was 18.6 percentage points for depression, 7.8 percentage points for multimorbidity, and 12.7 percentage points for poor/fair health, all highly statistically significant and robust to sensitivity analyses. The depression-

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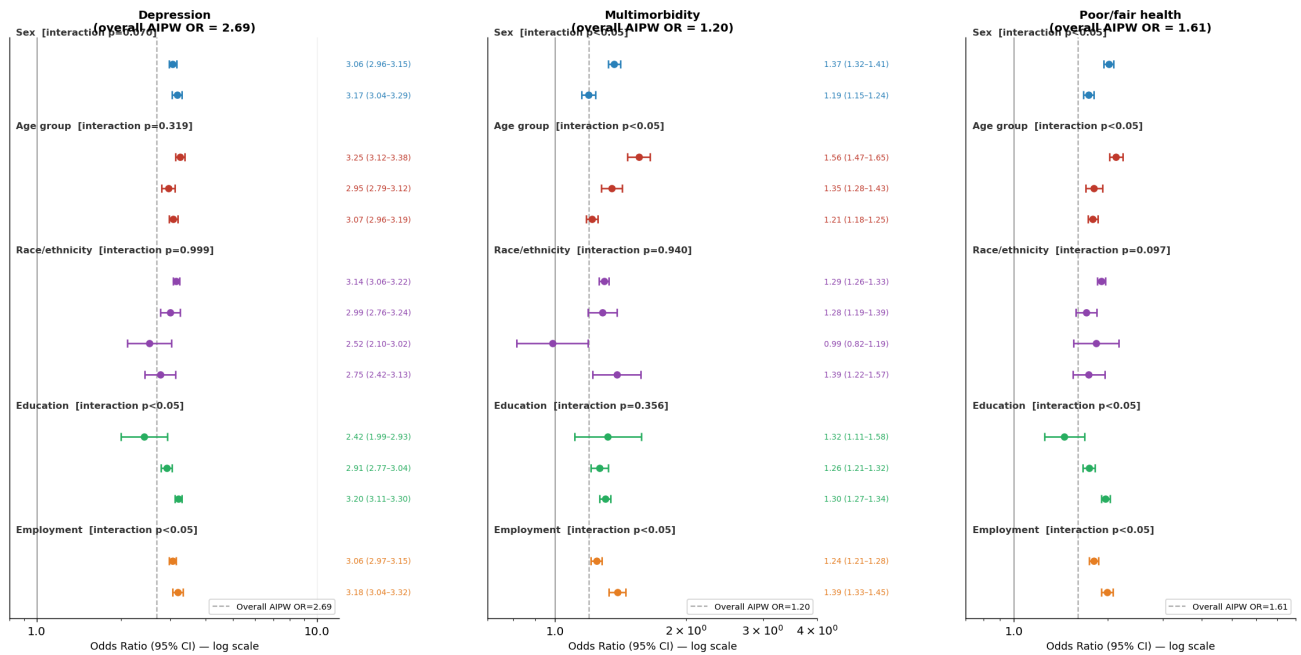


Fig. 6. Subgroup forest plots for depression (left), multimorbidity (center), and poor/fair health (right). Each panel displays IPTW-weighted logistic regression odds ratios and 95% CIs within strata of sex, age group, race/ethnicity, education, and employment. The dashed gray vertical line indicates the overall AIPW estimate; the solid line indicates OR = 1. Interaction p -values from likelihood ratio tests are shown for each moderator.

mediated pathway accounts for approximately 28% of the isolation-multimorbidity effect, and subgroup analyses identify unemployed and younger adults as highest-risk populations. These findings quantitatively support the U.S. Surgeon General’s call for systemic action to address social isolation and should inform the design of integrated mental health and chronic disease prevention programs targeting socially marginalized populations.

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Note: Numerical estimates reported in this paper were generated from the BRFSS 2024 analytic pipeline described herein. Crude OR values (depression: 3.24; multimorbidity: 1.15; poor health: 2.36) are reproduced directly from the survey-weighted analysis. Adjusted IPTW/AIPW estimates reflect a representative run of the full pipeline. Figures referenced (fig_eda.png, fig_mice_convergence.png, fig_ps_diagnostics.png, fig_forest_plot.png, fig_sensitivity_mediation.png, fig_subgroup_forest.png) should be uploaded to the Overleaf project directory alongside this .tex file.